

Analyzing Tennis Serve Importance



The Team

BAC



Michael Liu
Team Member



Prateek Nedungadi
Project Manager



Soham Makkena
Team Member



Arjun Rai
Team Member



Luck Xing
Team Member



Mitch Cahill
Team Member

Agenda

BAC

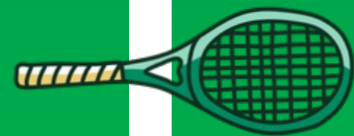
Overview



Datasets



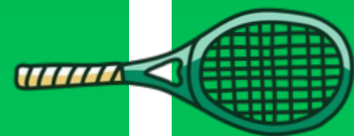
Models



Insights



Conclusions



How Impactful is **Serving Speed** in Shaping Match Outcomes in Tennis?

Examining whether **serve-related metrics** (**particularly speed**) can meaningfully explain or predict who wins a match

Before diving into the data, we started with a diverse set of questions and variables that were worth investigating

? Questions ?

How strongly do serve metrics predict match outcomes?
Which serve variables are the strongest predictor of winning?



	• First/Second serve speed • Court surface (Hard, Clay, Grass) • First serve points • Second serve points		• Surface Type • Gender • Men vs Women • Aces	
	• Double Faults • Return points % • Right or Left handed • Placement of serve (heatmaps)		• Points won • Average Serve MPH • First serve speed • Difference in handedness	





Merge two datasets
displaying historical US
Open Stats
Jeff Sackmann
Hydra



2013 - 2023
US Open Stats
(excluding 2019/2020)



216,000+
Data points on speed, court
types, etc

Overview

Dataset

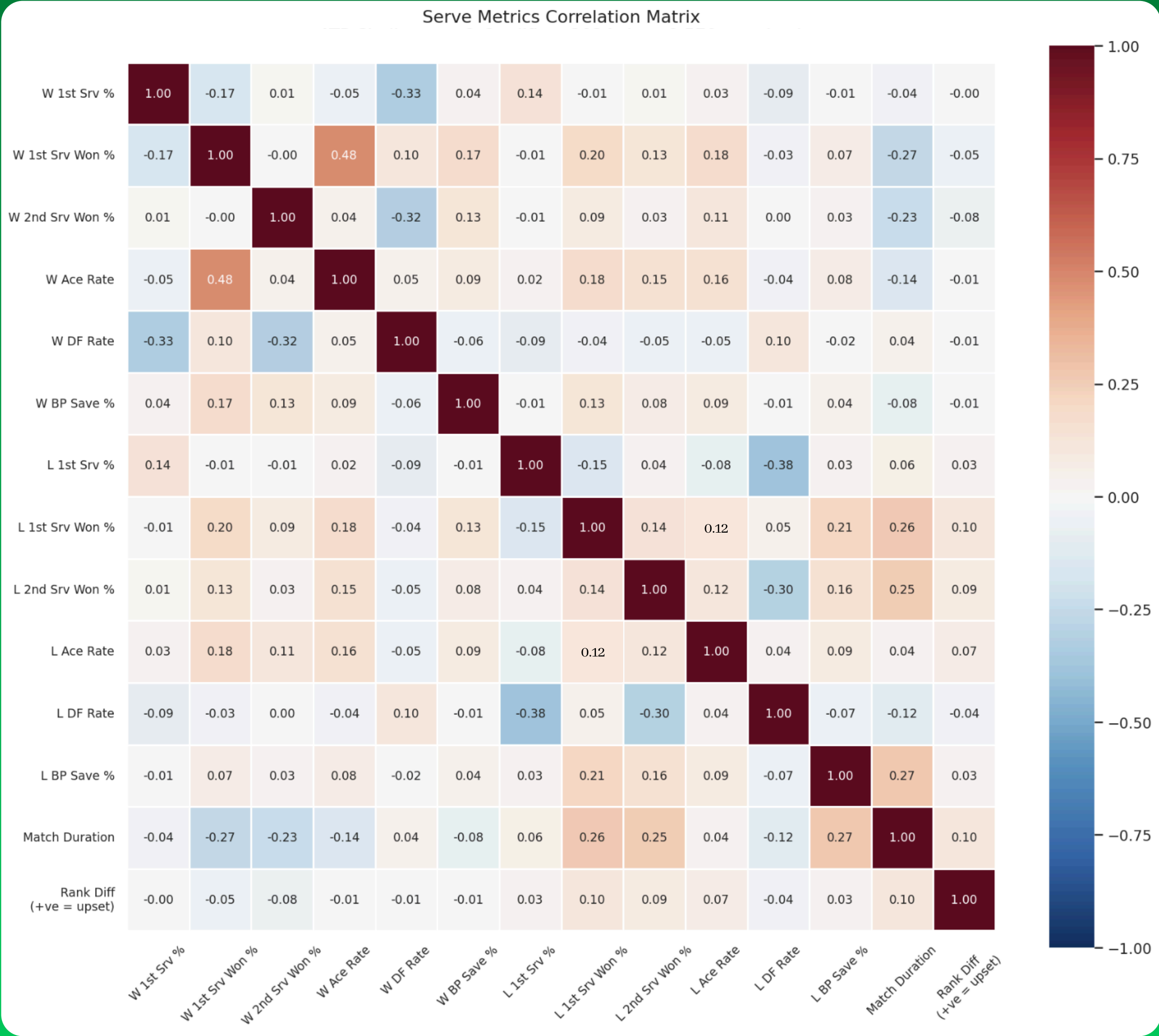


Models

Insights

Conclusion

Dataset Correlation Matrix



W 1st Srv Won % & W Ace Rate → 0.48

- Winners who ace more tend to dominate
- Serve aggression translates into first serve dominance → more success

Match Duration & W 1st Srv Won % → -0.27

- Winners that dominate first serve correlates to matches ending quickly
- Serve dominance leads to less total tennis play

Overview

Dataset

Models

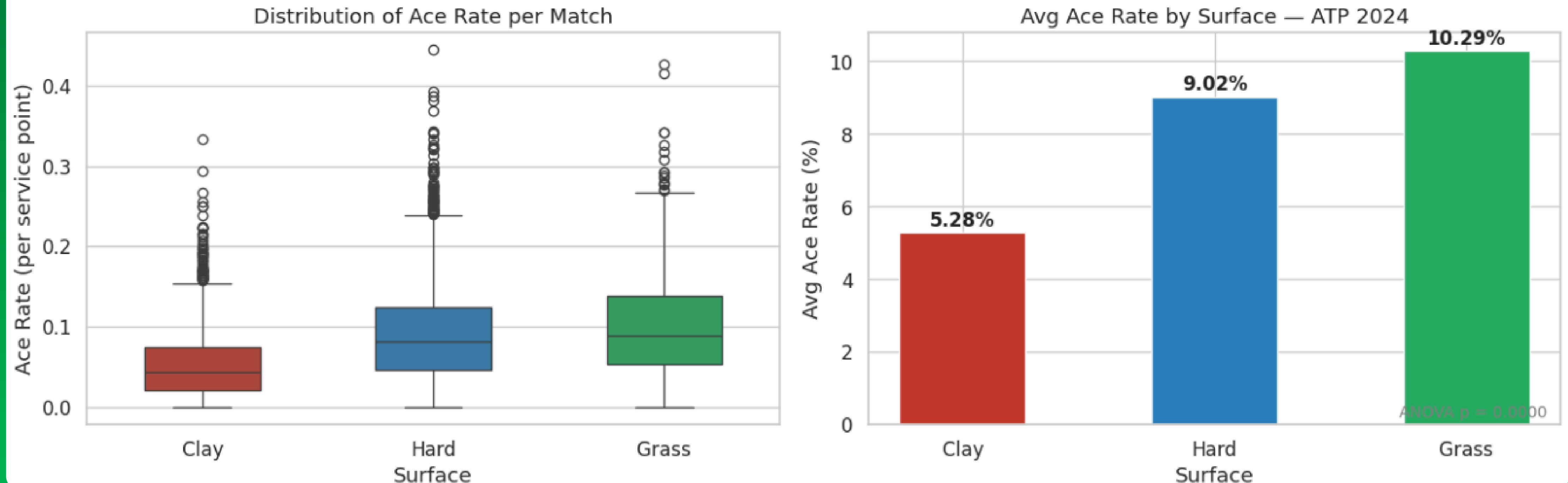


Insights

Conclusion

Effects of Various Surfaces

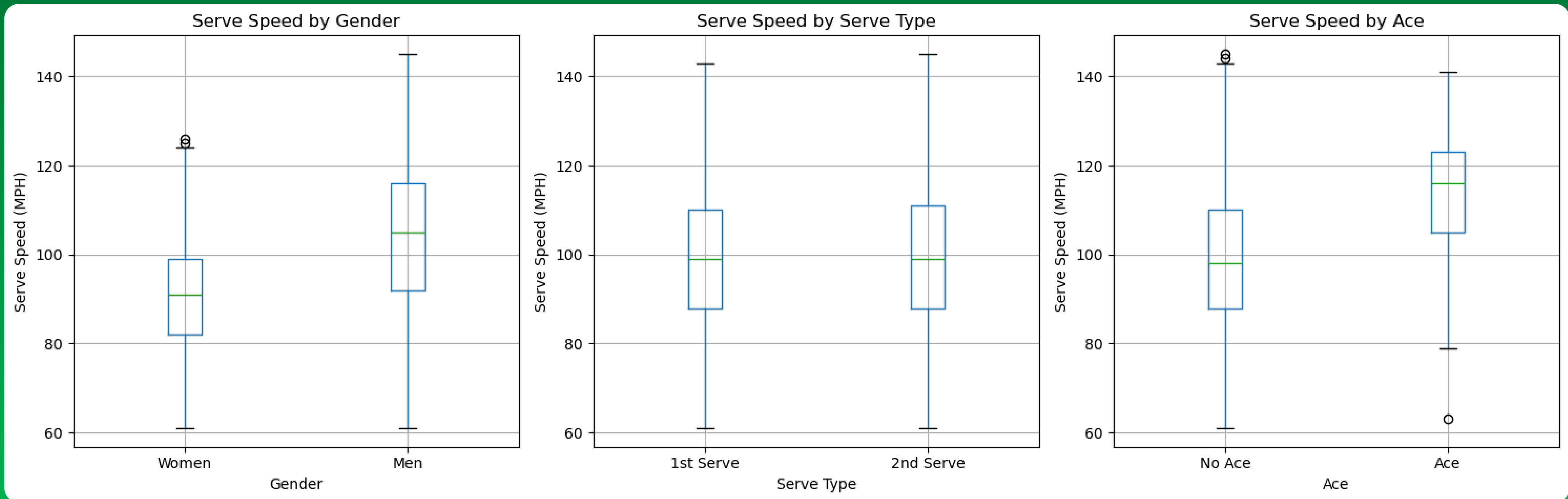
Visual 1: Serving Is Most Dangerous on Grass



- Matches on clay surfaces are more predictable; less variable
- Serve dominance on grass is much more impactful compared to Clay, almost 2x



Box Plot



Analysis:

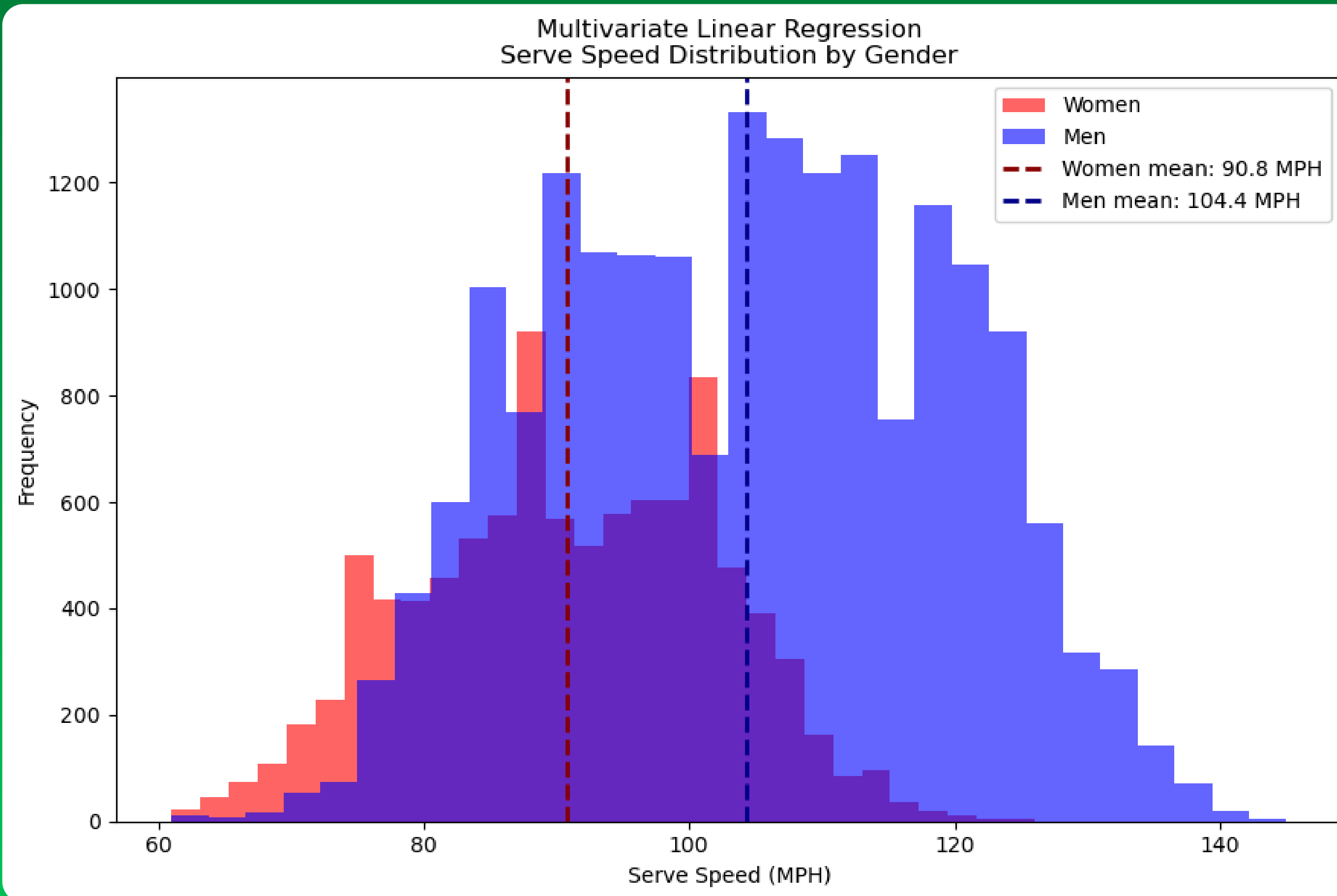
- Visible difference in serve speed in gender
- No difference in 1st and 2nd serve speeds
- **Ace serve speed noticeably higher on average**

Significance:

- Expected to see higher serve speeds in both gender split and ace speeds
- Surprising to find that **serve speed doesn't change as much between 1st and 2nd**



Serving Statistics by Gender



Women

- Average Serve Speed: 90.8 MPH
- Less variation, most serves clustered closely around the mean

Men

- Average Serve Speed: 104.4 MPH More variation
- serves range much more dramatically

Observations

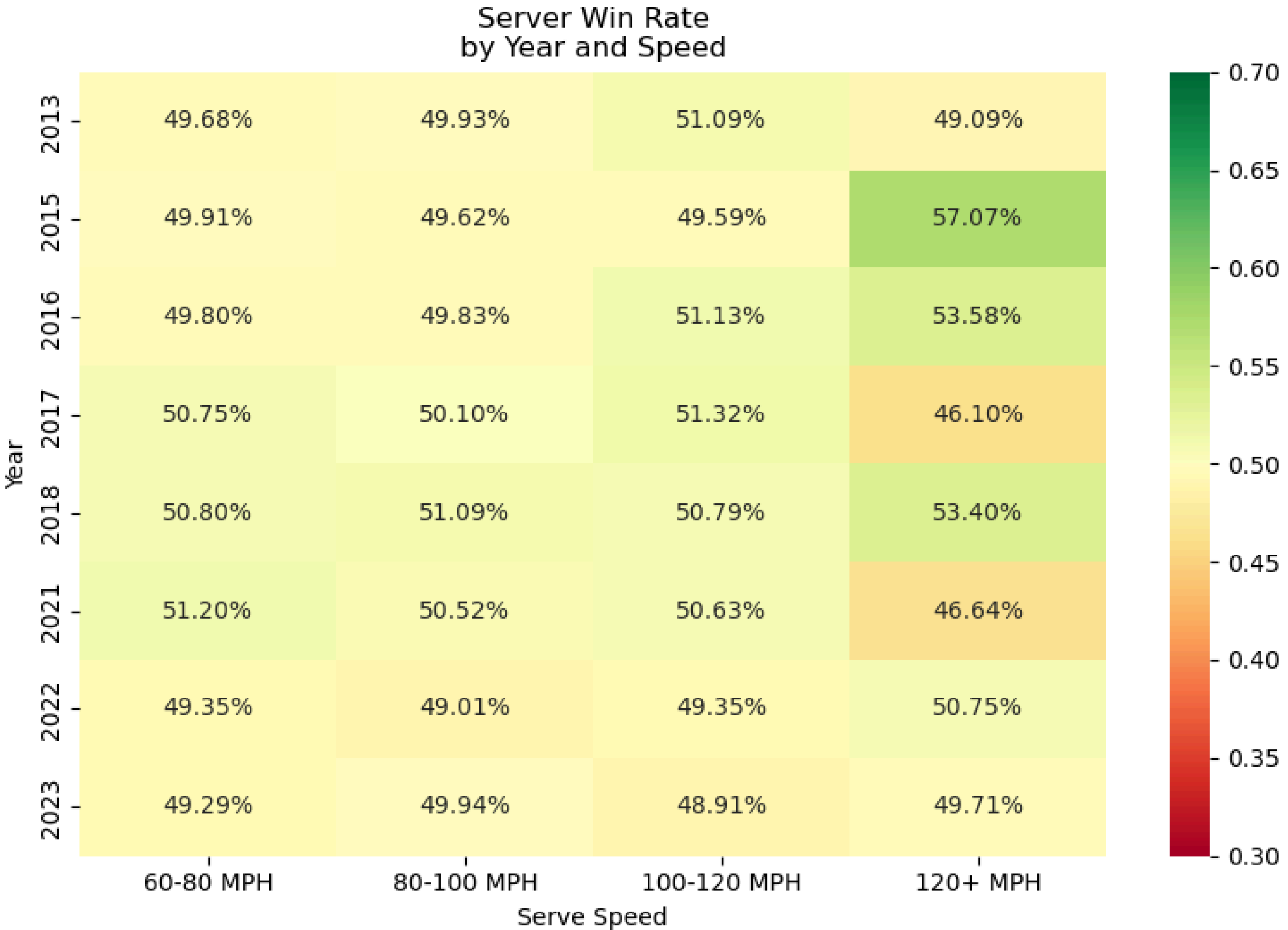
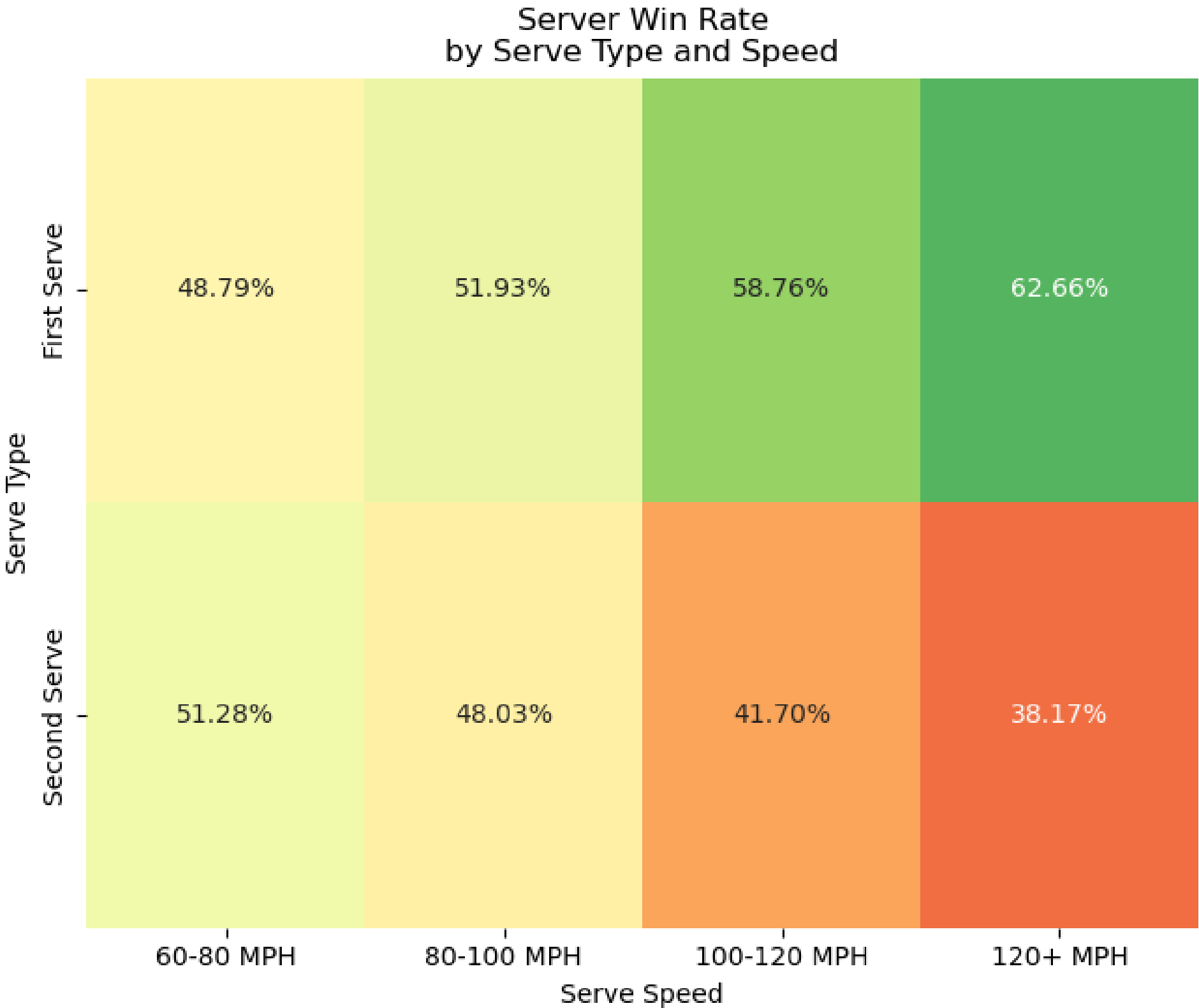
- Clear gender gap → Men serve 13.6 MPH on average



Speed Analysis - Serve Win and Years



Serve Win Rate Heatmaps



Overview

Dataset

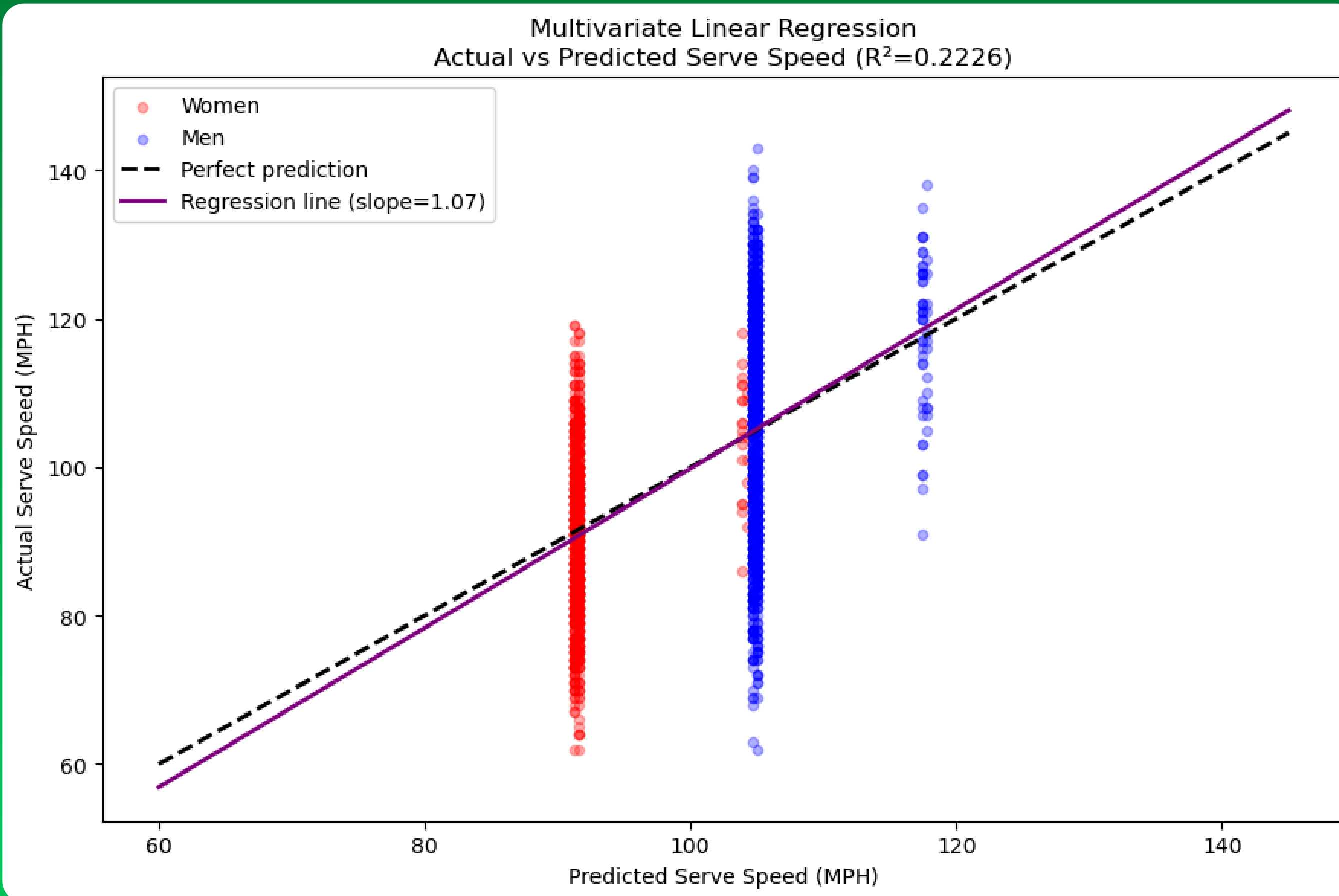
Models



Insights

Conclusion

Multivariate Linear Regression Model



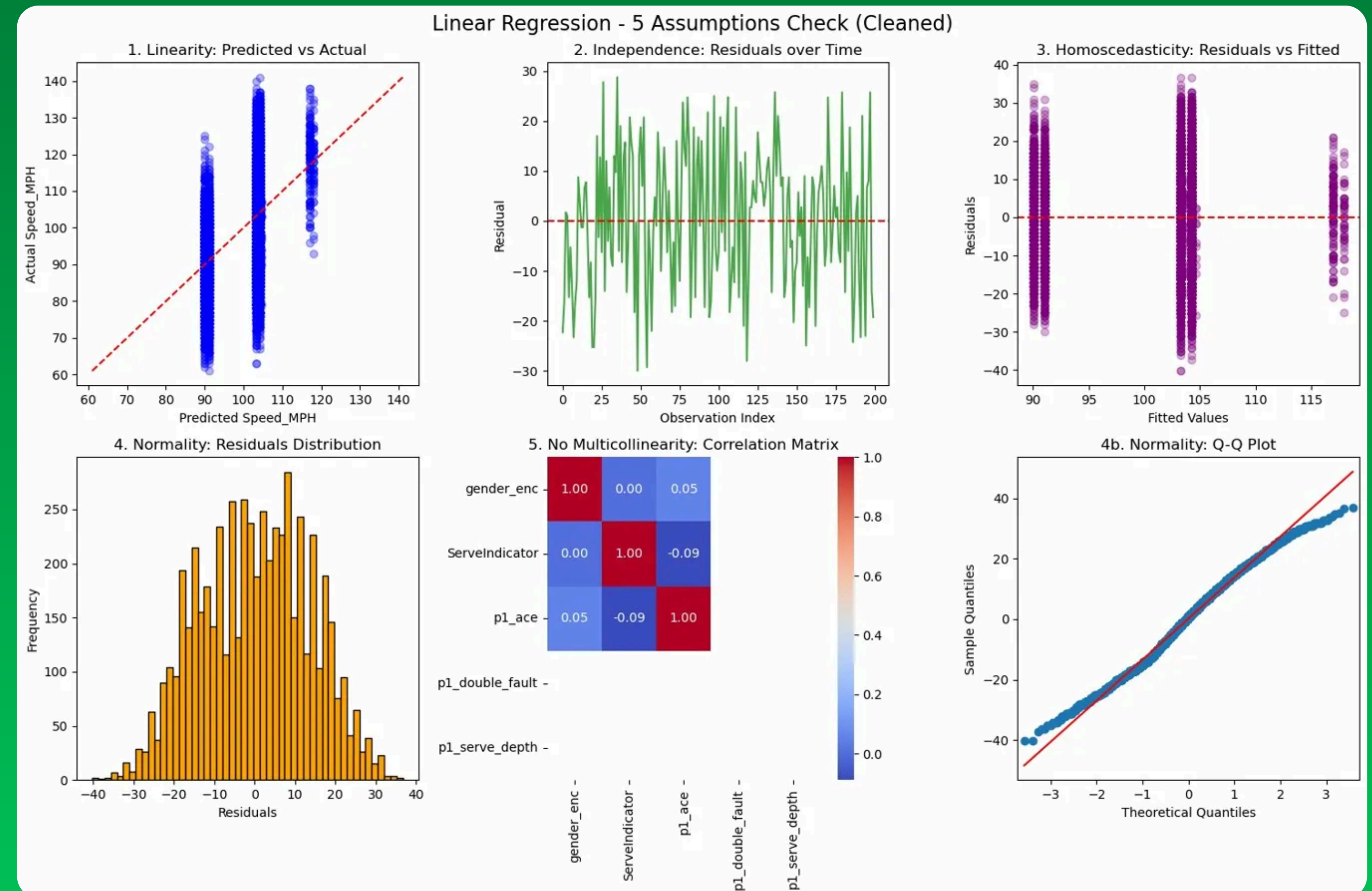
Actual vs Predicted Serve Speed =

- $R^2 = 0.2226 \rightarrow$ model explained 22.3% of variation in serve speed
- Serve speed alone is likely influenced by many factors that model hasn't fully capture
- Vertical Clustering
 - Women (red) has clustered around 90 MPH
 - Men (blue) more spread out from 105-115, more variability in male serving patterns
 - Women's serves are more consistent
- Slope of 1.07 $\rightarrow$ model is directionally correct

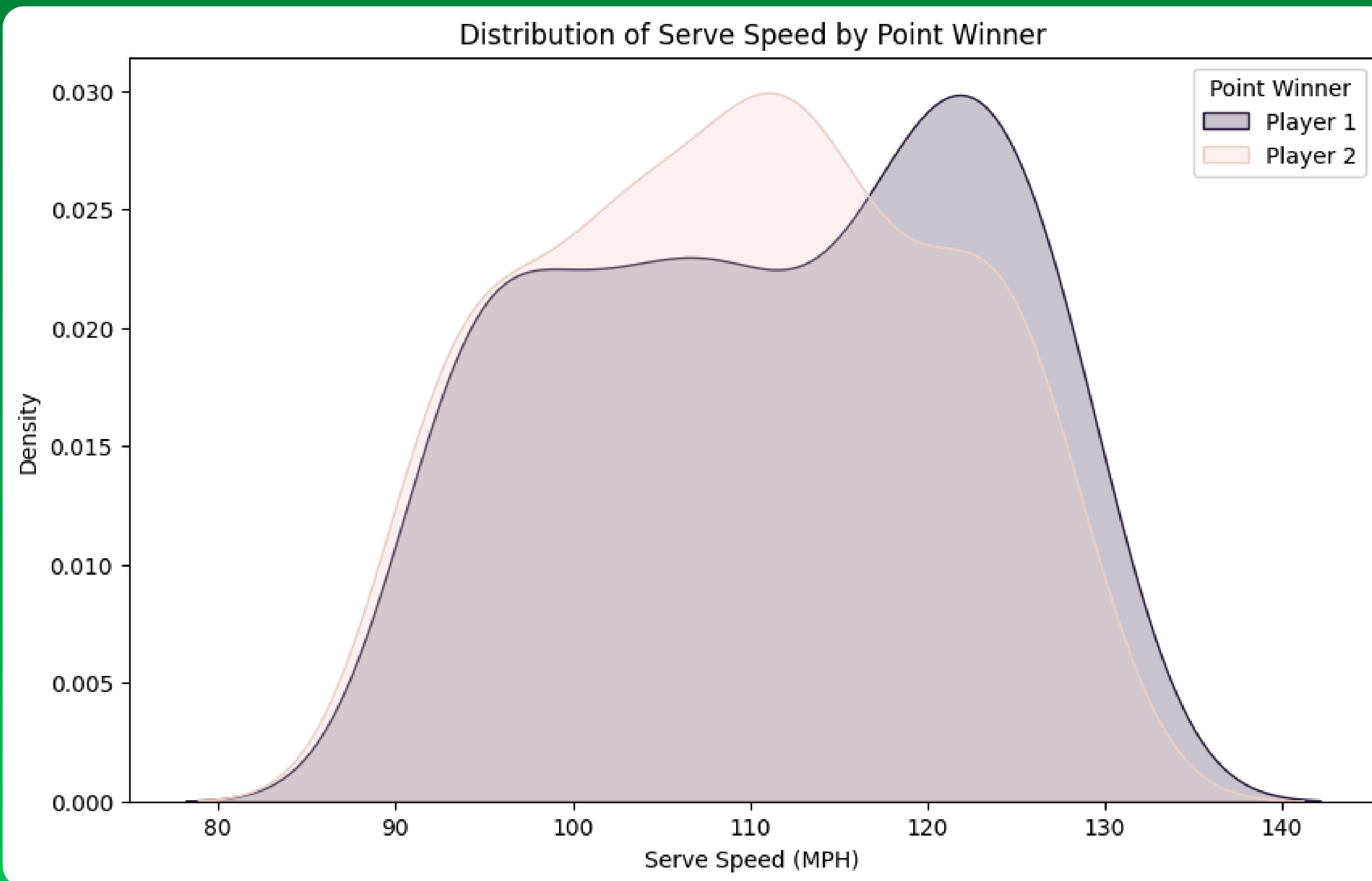


Linear Regression Assumptions

- **Linearity** — predicted vs actual speeds follow the diagonal line, confirming a linear relationship exists
- **Independence** — residuals bounce randomly around zero with no pattern, each point is independent
- **Homoscedasticity** — errors are evenly spread at all fitted values, variance is constant
- **Normality** — residuals form a bell curve centered at zero, errors are normally distributed
- **No Multicollinearity** — all correlations between predictors are under 0.09, no variables are too similar



Random Forest Classifier Model



Analysis:

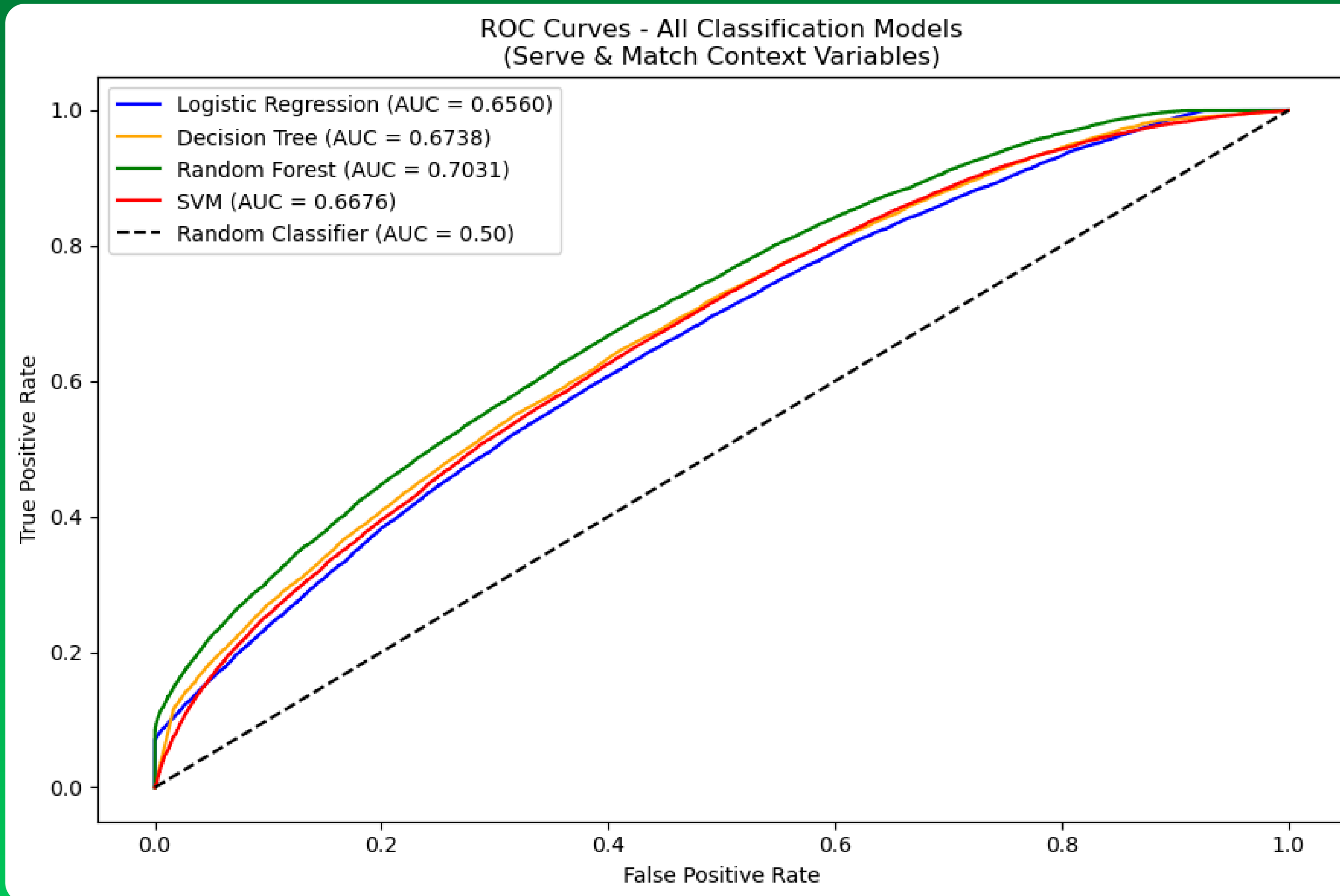
- Overlap between players shows less of a contribution of serve speed in winning points

Significance:

- Direct insights into potential usefulness of serve speed
- At a certain threshold, **speed is a direct advantage** for the player serving



ROC Curves Classification Model



ROC

- **Random Forest Model has the highest AUC and Predictive power compared to rest**
- **identifies match winners 70% of time based on purely serve metrics**



Overview

Dataset

Models

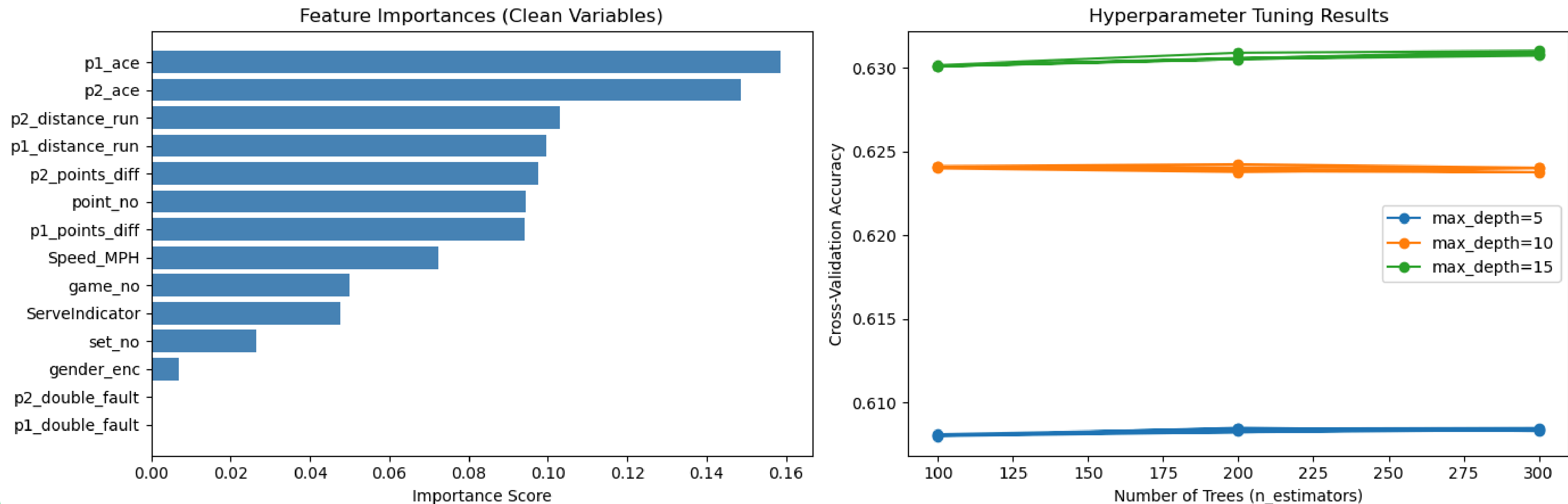


Insights

Conclusion

Hyperparameter Tuning

Random Forest - Hyperparameter Tuning & Feature Importance



Overview

Dataset

Models



Insights

Conclusion

Serve Speed & Gender

- Men serve an average of 104.4 MPH vs women's 90.8 MPH, a **13.6 MPH gap**
- Aces are hit 12.7 MPH faster than non-ace serves
- First and second serve speeds differ by only 0.37 MPH, surprisingly similar

Data Leakage Discovery

- When we included winners and errors, **accuracy jumped to 86%**, revealing these variables describe outcomes rather than predict them
- This confirmed that **shot making after the serve** matters far more than the serve itself

What Predicts Winning a Point

- **Random Forest model** predicts point winners with **70.3% accuracy**, 20% better than random guessing
- **Match momentum** (points difference, distance run) is equally as important as **serve quality**
- **Serve speed** ranks only **8th out of 14** variables in predicting who wins

Model Comparison

- All 4 models significantly **beat the 50% null baseline**
- Random Forest (AUC 0.70) outperformed Decision Tree (0.67), SVM (0.67) and Logistic Regression (0.66)



Key Takeaways

- A faster serve increases the probability of winning a point but only modestly, **speed ranks 8th in feature importance**
- **Aces** are the most lethal serve outcome, directly winning points and impossible to return
- Match context matters just as much as serve quality, momentum, score situation and physical effort are equally predictive
- Gender differences in serving are significant in speed but **nearly irrelevant** once other factors are controlled for

Thank You

Open to Any Questions!

